

Sub-critical crack growth rate of soda-lime-silicate glass and less brittle glass as a function of fictive temperature

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Abstract

Sub-critical crack growth rates of soda-lime-silicate glass and less brittle glass with different fictive temperatures were compared using the DCDC method under both dry and humid atmospheres in order to investigate the origin of the unique mechanical features of the less brittle glass developed by Ito and his collaborators. In both dry and humid atmospheres, the crack velocity of the soda-lime-silicate glass was slower than that of the less brittle glass. For both glasses, the glass sample with higher fictive temperature showed a slower crack growth rate under both dry and humid atmospheres. These observations can be explained by the tendency for the plastic flow at the crack tip; the soda-lime-silicate glass is expected to show easier plastic flow under tension than the less brittle glass, and also the samples with higher fictive temperatures are expected to show easier plastic flow, leading to greater fracture toughness, K_{IC} , and slower crack growth rate.
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1. Introduction

Inorganic glasses are considered purely brittle materials and their strength is greatly affected by the size of surface flaws. For example, the strength of silica glass fiber without any surface flaws, i.e., pristine fiber is approximately 10–15 GPa at liquid nitrogen temperature [1] and 5–6 GPa at room temperature [2], while the strength of bulk silica glass is approximately two orders of magnitude lower. The difference is due to the surface flaws on the bulk glass. The flaws develop easily on the glass surface because of the brittle nature of the glass. It is a great challenge to develop a less brittle glass with less propensity to form the surface flaws. Brittleness was defined by Lawn and Marshall [3] using the parameter $B = H/K_C$, where H is hardness and K_C is fracture toughness. The hardness represents resis-

tance to deformation and the fracture toughness represents resistance to fracture. Thus, a glass with easier plastic deformation and higher toughness is expected to exhibit a lower brittleness.

Sehgal and Ito [4] investigated the brittleness of various glasses in a silicate system and developed a unique ‘less brittle glass’. The glass has a similar glass composition to the commercial window glass yet it exhibits remarkable features such as greater crack initiation load under indentation and cleaner scratch patterns. The glass composition and some properties of the less brittle glass are listed in Table 1 together with those of a typical soda-lime-silicate glass. Ito and his collaborators analyzed the less brittle glass in comparison with the soda-lime-silicate glass, by various means including molecular dynamics calculations [5,6].

The development of the less brittle glass gives us a hope to create much less brittle glasses. However, in order to realize this dream, one has to understand the origin of the less brittleness of the developed glass. Here, slow crack

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Table 1
The glass composition and properties of the soda-lime-silicate glass and the less brittle glass

	Composition (mol%)						Density (g/cm ³)	Young's modulus (GPa)	T_g (°C)
	SiO ₂	Al ₂ O ₃	Na ₂ O	K ₂ O	MgO	CaO			
Soda-lime-silicate glass	71.7	1.1	12.2	0.6	5.6	8.8	2.5	72	550
Less brittle glass	78.6	2.0	13.3	1.1	4.0	1.0	2.4	68	515

growth rates of less brittle glass are compared with those of the soda-lime-silicate glass.

Properties of glasses generally vary with their cooling rate. This phenomenon is attributed to the fact that glasses acquire different fictive temperatures [7] when cooled at different rates; a sample cooled more rapidly has a higher fictive temperature. A soda-lime-silicate glass sample cooled at a faster rate i.e., a sample with higher fictive temperature, shows lower density [8], lower hardness [9] and lower Young's modulus [10].

Correspondingly, mechanical strength of glass is expected to change with fictive temperature. Earlier, it was reported that soda-lime-silicate glass with higher fictive temperature exhibited higher strength [11] and greater fatigue resistance [12] as well as a slower crack growth rate both in dry and wet atmospheres [13]. Ito et al. [9] investigated the fictive temperature dependence of brittleness and fracture toughness of the soda-lime-silicate glass and the less brittle glass by Vickers indentation and found that soda-lime glass with a higher fictive temperature had a lower brittleness and higher fracture toughness while less brittle glass did not exhibit much dependence on fictive temperature.

Measurement of mechanical properties of a glass with a different fictive temperature helps explain the origin of some phenomenon such as brittleness, since the change of chemical composition, which could complicate the interpretation of the phenomenon, is not involved. Mechanical properties such as fatigue resistance and fracture toughness can be determined by measuring the relationship between stress intensity (K_I) and crack velocity (V), the so-called K_I - V curve, as shown schematically in Fig. 1. Generally, the K_I - V curves are divided into three regions, region I, II and III. In region I, crack velocity increases with applied stress as well as with humidity. Generally, this is explained as crack growth being limited by the rate of reaction between Si-O-Si bond and water at the crack tip. The crack growth rate in this region is closely related to mechanical fatigue of the glass. In region II, crack velocity is independent of applied stress but depends on humidity. This is explained as crack growth being limited by the rate of transport of water vapor to the crack tip. Thus, the crack growth rates in region I and region II are dependent on humidity. On the other hand, crack growth rate in region III is independent of humidity but changes sharply with the applied stress. In a water-free environment, or in a vacuum, only region III should be observed. The critical stress intensity, K_{IC} , which is the fracture toughness, is commonly defined as the value of K_I required for crack

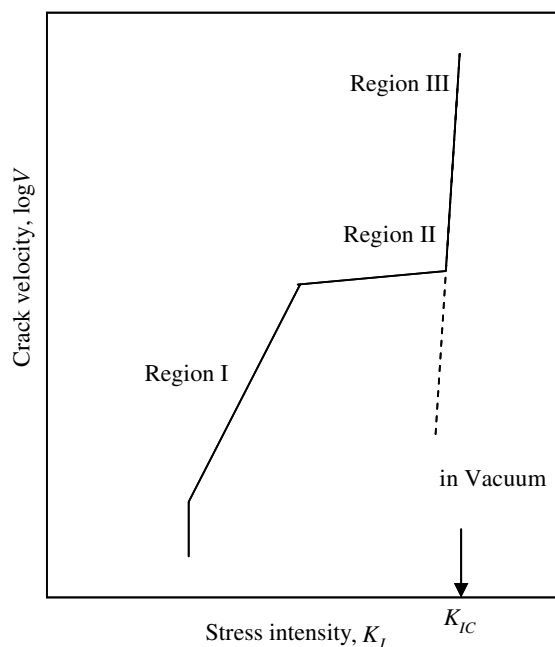


Fig. 1. The schematic of the K_I - V curve for soda-lime-silicate glasses. K_{IC} is the critical stress intensity, i.e., fracture toughness. The dashed line shows the K_I - V curve in a vacuum.

growth in an inert atmosphere. The crack growth in region III reflects intrinsic crack growth, i.e., $K_I = K_{IC}$.

In this study, sub-critical crack growth rates of less brittle glass with different fictive temperatures were measured using the DCDC fracture mechanics technique under both dry and humid atmosphere and compared with the similar data obtained for soda-lime-silicate glass.

2. Experimental procedure

The less brittle glass developed by Ito et al. was used in this study. Slow crack growth data for a commercial soda-lime-silicate glass (AS, a product of Asahi Glass Co., Ltd., Japan) obtained earlier [13] was used for comparison. Nominal compositions and some selected properties of both glasses are shown in Table 1. The glass samples for the DCDC measurement were fabricated into bars with the dimensions of 75 mm × 7.5 mm × 6.5 mm. The sample surfaces were polished by using a cerium oxide slurry so that one can observe the crack growth from the outside of the glass sample. A hole 1 mm in radius was core-drilled through each sample at the center of the 75 mm × 7.5 mm

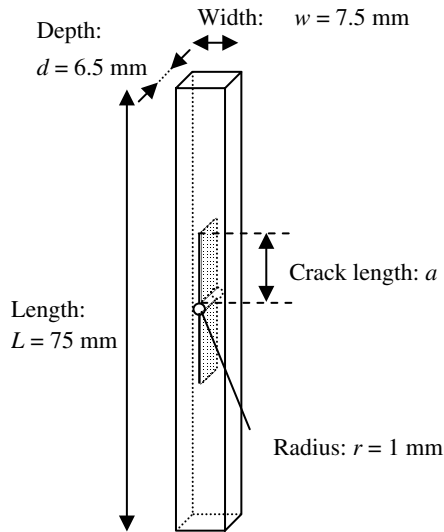


Fig. 2. The DCDC specimen geometry employed in this study. Two cracks initiate from the hole and extend along the long axis of the specimen under compressive stress.

face of the specimen. This is the same geometry as that used by Michalske et al. [14] and is shown in Fig. 2.

Glass samples were heat-treated at various temperatures to produce samples with different fictive temperatures. The soda-lime-silicate glass samples were heated at 450 °C for 300 h or 550 °C for 16 h, and the less brittle glass samples were heated at 415 °C for 300 h or 515 °C for 16 h in order to prepare samples with different fictive temperatures. The experiment was designed to test samples of both glasses having the fictive temperature difference of 100 °C. Dry nitrogen gas with relative humidity below 0.03% was passed through the furnace during heat treatment. The fictive temperatures of the soda-lime-silicate glass were measured by a simple IR method [15] as described earlier. It was confirmed that the fictive temperatures on the fracture surface of the soda-lime-silicate glass specimen was the same as the heating temperatures within the measuring error of ± 10 °C. Furthermore, the fictive temperatures of both the soda-lime-silicate glass samples and the less brittle glass samples were examined by measuring the sample density. The fictive temperature coefficient of the density of the soda-lime-silicate glass and the less brittle glass are $0.00015 \text{ (g/cm}^3\text{)/}^\circ\text{C}$ and $0.00010 \text{ (g/cm}^3\text{)/}^\circ\text{C}$, respectively [8,16]. The density difference between high and low fictive temperature samples was slightly smaller than expected for both samples. Therefore, the fictive temperatures in this experiment may have some error; the lower fictive temperature samples especially may have 10–20 °C higher fictive temperatures than the fictive temperatures shown here.

The samples were loaded in compression in an Instron Model 8562 mechanical test machine with 5000 kg load cell. The temperature was kept at 22 ± 1 °C. For the wet condition, ambient air with a relative humidity of approximately 50% was used. For the dry condition, dry nitrogen with a relative humidity below 0.03% was used. Two starter

cracks were cut at the top and the bottom of the hole in the specimen using 0.3-mm-diameter diamond-coated wire. Two stainless steel hemispheres of 6.5 mm-diameter were attached to the both ends of the specimen. The load was increased in 10–20 kg increments until the crack growth was observed by using a cathetometer with precision of 10 μm . Crack position was measured at the center of the specimen. When the crack reached around 4–5 mm and the crack growth rate became too slow to measure, the load was raised by an additional 20 kg and then the crack growth rate increased, enabling one to measure the crack growth rate again. Initially, the crack grew rapidly under a constant load and the crack lengths were measured in intervals as short as 15 s. As the crack velocity slowed, the measurement intervals between readings were increased proportionally.

The data were analyzed to obtain stress intensity (K_I) and crack velocity (V). The stress intensity for a given crack length is obtained by He et al. [17] for the sample geometry employed in this experiment

$$K_I = \frac{\sigma \sqrt{\pi r}}{(w/2r) + \frac{\{0.235w/2r - 0.259\}a}{r}}, \quad (1)$$

where σ is the applied stress, r is the hole radius, w is the specimen width and a is the crack length measured from the hole surface. The results from this equation are accurate within 1% for the crack lengths in the range $w/2r \leq a/r \leq 15$. In this study, Eq. (1) was used for the calculation of K_I .

3. Results

Fig. 3 reproduces the K_I – V curves, obtained earlier [13], for the soda-lime-silicate glasses with different fictive temperatures in the ambient air with relative humidity around 50% and with the dry nitrogen atmosphere. The lines were obtained by fitting the power-law expression

$$V = AK_I^n \quad (2)$$

to the data obtained in the dry atmosphere, where A and n are constants and n is called the stress-corrosion susceptibility coefficient. The values of n were 19.1 for the high fictive temperature sample and 21.2 for the low fictive temperature sample in region I and approximately 140 ± 10 for both specimens in region III. The data obtained in the wet atmosphere correspond to region I and those in the dry atmosphere correspond to region III. In the wet atmosphere, the soda-lime-silicate glass with higher fictive temperatures showed slower crack growth. In the dry atmosphere, there was little difference in the crack growth rates of glasses with different fictive temperatures, which is less than experimental error. The error range shown in Fig. 3, was mainly caused by the measuring error of the hole radius. For example, in the case of $a = 7.00$ (mm), $w = 7.50$ (mm) and $r = 1.000$ (mm), using the measuring error of 20 μm in hole radius, r , which corresponds to 2%

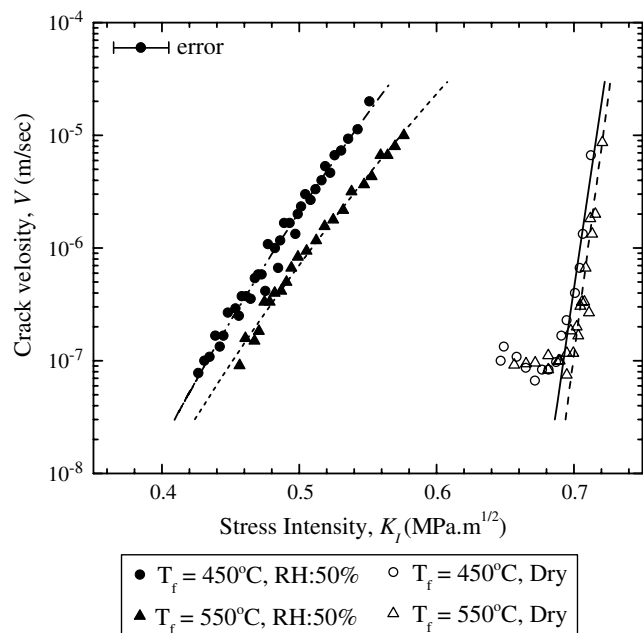


Fig. 3. The K_I - V curve for the soda-lime-silicate glasses with different fictive temperatures in the ambient air with relative humidity around 50% and in a dry nitrogen atmosphere. The solid line and dashed line in region III were obtained by fitting Eq. (2) to 450 °C fictive temperature data with $n = 134$ and 550 °C fictive temperature data with $n = 150$, respectively. Dash-dotted line and short dashed line in region I were obtained by fitting the Eq. (2) to 450 °C fictive temperature data with $n = 21.2$ and to 550 °C fictive temperature data with $n = 19.1$, respectively. The error range in the crack velocity under constant stress intensity is approximately the same as the size of the symbols.

of the nominal radius, the error around 4.5% was estimated for the K_I value by Eq. (1). However, since the hole was drilled using the same diamond drill, the differences of the hole sizes between different samples with different fictive temperatures of the same glass are expected to be small. Therefore, although the obtained data could shift in terms of the absolute value of K_I , the relative positions of the data for the samples with different fictive temperatures would not be affected. Thus, it can be concluded that a similar tendency as observed in the wet atmosphere was also observed in the dry atmosphere. The error range in the crack growth rate at a given stress intensity was approximately the same as the size of the symbols in Fig. 3.

Fig. 4 shows the K_I - V curve for the less brittle glasses with different fictive temperatures in the ambient air with relative humidity of approximately 50% and the dry nitrogen atmosphere. For comparison, the slow crack growth data for the soda-lime-silicate glass are shown as dotted line in this figure. The values of n were 23.2 for the high fictive temperature sample and 23.6 for the low fictive temperature sample in region I and 150 ± 10 for both specimens in region III. In the wet atmosphere, the less brittle glass with higher fictive temperatures showed slower crack growth. In the dry atmosphere, the difference between the high fictive temperature sample and the low fictive temperature sample was small, but a similar tendency as in the wet atmosphere

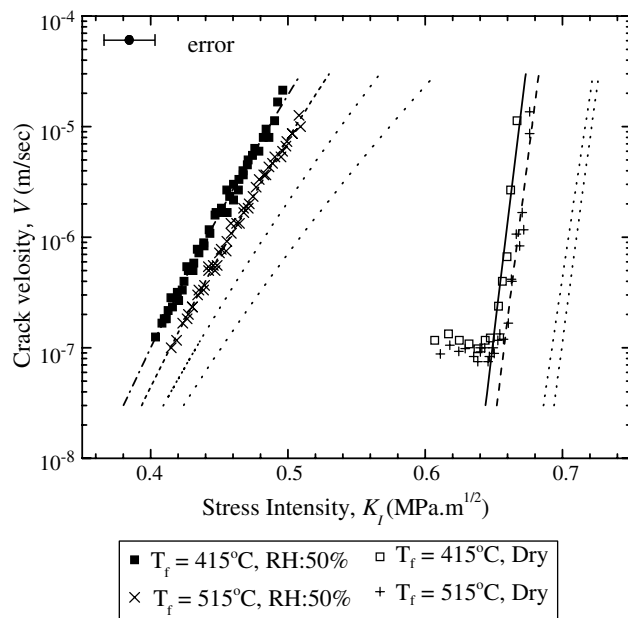


Fig. 4. The K_I - V curve for the less brittle glasses with different fictive temperatures in the ambient air with relative humidity around 50% and in the dry nitrogen atmosphere. The solid line and dashed line in region III were obtained by fitting Eq. (2) to 415 °C fictive temperature data with $n = 155$ and 550 °C fictive temperature data with $n = 150$, respectively. The dash-dotted line and short dashed line in region I were obtained by fitting Eq. (2) to 415 °C fictive temperature data with $n = 23.6$ and to 515 °C fictive temperature data with $n = 23.2$, respectively. The dotted lines without data points represent the soda-lime-silicate data from Fig. 3. The error range in the crack velocity under constant stress intensity is approximately the same as the size of the symbols.

was observed; the glass with the higher fictive temperature showed a slower crack growth rate. These results imply that both the soda-lime-silicate glass and the less brittle glass with higher fictive temperature should show higher strength. The present observation appears to be consistent with the previous observations in the dynamic fatigue experiment, where the glass samples with higher fictive temperature exhibited a greater strength [11,12].

4. Discussion

Fictive temperature dependences of fracture toughness and brittleness in the soda-lime-silicate glass and the less brittle glass were investigated by Ito et al. [9] using an indentation method. Their data showed that the fracture toughness, K_{IC} , of the less brittle glass was higher than that of the soda-lime-silicate glass. In addition, the K_{IC} of the soda-lime-silicate glass increased with increasing fictive temperature while that of the less brittle glass decreased with increasing fictive temperature. In the present results of the DCDC experiments, however, the fracture toughness, K_{IC} , of the less brittle glass was smaller than that of the soda-lime-silicate glass, and K_{IC} of the less brittle glass increased with increasing the fictive temperature as was the case for the soda-lime-silicate glass.

Ito [6] also investigated the fracture behavior of the soda-lime-silicate glass and the less brittle glass by molecular dynamics simulation. He found that both glasses showed plastic flow under tension while they showed both plastic flow and densification under compression. In addition, he suggested that the plastic flow in the soda-lime-silicate glass was easier than that in the less brittle glass since the less brittle glass has a higher polymerized network structure than soda-lime-silicate glass. On the other hand, the densification under an indenter in the less brittle glass, which contains a larger concentration of silica, is expected to be larger than that in the soda-lime-silicate glass, which contains a smaller concentration of silica. Bridgman and Simon [18] showed that the densification under the compression increased with silica content in sodium silicate glasses. Correspondingly, indentation-induced densification data obtained by Yoshida et al. [19] showed glasses with higher silica content tend to exhibit greater densification. In the indentation method, fracture toughness, K_C , is measured under a compressive loading. Therefore, both plastic flow and densification affect the obtained K_C . However, in the DCDC experiment, the K_I – V curve is measured under tension so that densification does not take place.

Plastic deformation such as plastic flow can affect crack propagation at the crack tip. The energy release rate can be calculated from the stress intensity, K_I , and Young's modulus, E , as

$$G_I = K_I^2/E \quad (3)$$

and represents the energy change due to the crack propagation. In the Irwin–Orowan model [20,21], the critical energy release rate, G_C , can be expressed by crack resistance energy, R

$$G_C = K_{IC}^2/E = R \quad (4)$$

and R can also be expressed by

$$R = 2\gamma + R_P, \quad (5)$$

where γ is the fracture surface energy and R_P is the plastic work term [22]. Thus, easier plastic deformation tends to increase crack resistance energy, R , and also K_{IC} as well as G_C . Thus, when the plastic deformation takes place at the crack tip, a greater fracture surface energy is required and the crack growth rate becomes smaller under a given stress intensity. The higher K_{IC} of the soda-lime-silicate glass by the DCDC experiment appears to be attributable to easier plastic flow.

On the other hand, it is expected that the fracture toughness, K_C , measured by the indentation will be influenced by the extent of densification. The observed greater stress intensity of less brittle glass measured by the indentation method can be attributed to the greater extent of densification of this glass.

The effect of the cooling rate on the fracture process in soda-lime-silicate glass was investigated by Ito and Taniguchi [23,24] using molecular dynamics simulation. They showed that a quenched soda-lime-silicate glass, which

had a higher fictive temperature, exhibited easier plastic flow and a higher self diffusion coefficient of ions such as sodium ions compared with an annealed glass. This can cause higher fracture toughness, and this is consistent with the present results of the soda-lime-silicate glass. The less brittle glass is also expected to show plastic flow but its extent of plastic deformation is expected to be less than that of the soda-lime-silicate glass. It appears that easier plastic flow in the sample with the higher fictive temperature produces higher fracture toughness for less brittle glass, also. These expectations are consistent with the observed crack growth rates.

The brittleness obtained from the indentation experiment represents the resistivity to cracking by contact stress well. In fact, the crack initiation load of the less brittle glass measured by the Vickers indentation was almost 10 times higher than that of the soda-lime-silicate glass [4]. However, the present observation indicates that once a crack is initiated in the less brittle glass, this crack can propagate relatively easily. The unique feature of the less brittle glass appears to originate primarily from its ability to reduce the tensile stress imparted by contact with a hard object such as indenter or cutter. If a glass shows both lower brittleness obtained by indentation and higher K_{IC} obtained by the DCDC experiment, such a glass would be a truly tough glass.

5. Conclusions

Sub-critical crack growth rates of the soda-lime-silicate glass and the less brittle glass with different fictive temperatures were compared using the DCDC method under both dry and humid atmospheres. The soda-lime-silicate glass had a slower crack growth rate than the less brittle glass. In both glasses, the glass with higher fictive temperature showed a slower crack growth rate. The soda-lime-silicate glass with a higher fictive temperature is expected to show easier plastic flow leading to a higher fracture toughness. The less brittle glass is also expected to show plastic flow under tension. The slower crack growth in the less brittle glass with higher fictive temperature appeared to be caused by easier plastic flow, too. In addition, the K_{IC} of the less brittle glass was smaller than that of the soda-lime-silicate glass, while the K_C obtained earlier by an indentation method was higher than that of the soda-lime-silicate glass. Since the indentation involves compressive stress, both plastic flow and densification can occur, affecting the obtained K_C values. On the other hand, K_{IC} obtained by the DCDC experiment was measured under tension so that only plastic flow affected the obtained K_{IC} values. It was expected that the less brittle glass tended to show larger densification under compression but smaller plastic flow under tension than the soda-lime-silicate glass. Therefore, the less brittle glass showed lower K_{IC} in this experiment. These observations imply that a crack is difficult to initiate in the less brittle glass under indentation but the initiated crack can propagate relatively easily under a tensile stress.

References

- [1] B.A. Proctor, I. Whitney, J.W. Johnson, *Proc. Roy. Soc. A* 297 (1967) 534.
- [2] C.R. Kurkjian, P.K. Gupta, R.K. Brow, N. Lower, *J. Non-Cryst. Solids* 316 (2003) 114.
- [3] B.R. Lawn, D.B. Marshall, *J. Am. Ceram. Soc.* 62 (1979) 347.
- [4] J. Sehgal, S. Ito, *J. Am. Ceram. Soc.* 81 (1998) 2485.
- [5] J. Sehgal, S. Ito, *J. Non-Cryst. Solids* 253 (1999) 126.
- [6] S. Ito, *J. Ceram. Soc. Jpn.* 112 (2004) 477.
- [7] A.Q. Tool, *J. Am. Ceram. Soc.* 29 (1946) 240.
- [8] M. Hara, S. Suetoshi, *Rep. Res. Lab. Asahi Glass Co., Ltd.*, 5 (1955) 126.
- [9] S. Ito, J. Sehgal, S. Deutschbein, in: *Proceedings of the ICG Annual Meeting 2000*, Amsterdam, 2000, p. S1.1.
- [10] G.E. Stong, *J. Am. Ceram. Soc.* 20 (1937) 16.
- [11] B. Varughese, Y.-K. Lee, M. Tomozawa, *J. Non-Cryst. Solids* 241 (1998) 134.
- [12] H. Li, A. Agarwal, M. Tomozawa, *J. Am. Ceram. Soc.* 78 (1995) 1393.
- [13] A. Koike, M. Tomozawa, *J. Non-Cryst. Solids* 352 (2006) 5522.
- [14] T.A. Michalske, W.L. Smith, J.E. Houston, *J. Appl. Phys.* 76 (1994) 2109.
- [15] A. Agarwal, M. Tomozawa, *J. Am. Ceram. Soc.* 78 (1995) 827.
- [16] S. Ito, Asahi Glass Co., Ltd., unpublished.
- [17] M.Y. He, M.R. Turner, A.G. Evans, *Acta Metall. Mater.* 43 (1995) 3453.
- [18] P.W. Bridgman, I. Simon, *J. Appl. Phys.* 24 (1953) 405.
- [19] S. Yoshida, J.-C. Sangleboeuf, T. Rouxel, *J. Mater. Res.* 20 (2005) 3404.
- [20] G.R. Irwin, *Fracture dynamics*, in: *Fracturing of Metals*, American Society for Metals, Cleveland, OH, 1948, p. 147.
- [21] E. Orowan, *Rep. Prog. Phys.* 12 (1948) 185.
- [22] B.R. Lawn, *Fracture of Brittle Solids*, 2nd Ed., Cambridge University, Cambridge (UK), 1993.
- [23] S. Ito, T. Taniguchi, *J. Non-Cryst. Solids* 349 (2004) 173.
- [24] T. Taniguchi, S. Ito, *Phys. Chem. Glasses* 45 (2004) 184.